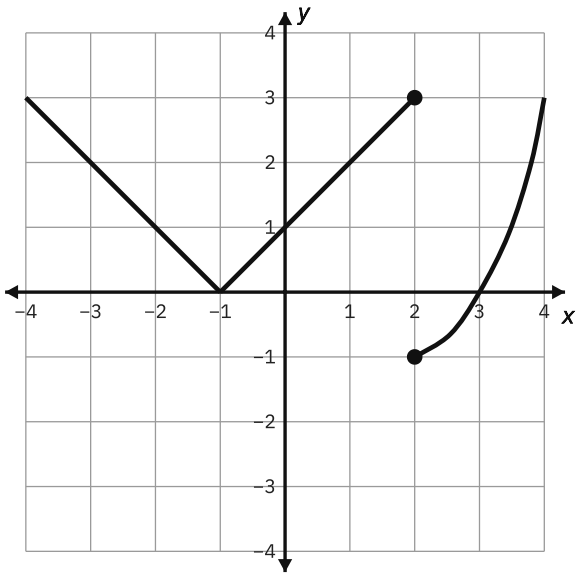
STATION

1

UNIT 5 REVIEW #2

Is it a function?

Marcel claims that the graph below represents a function.



State whether Marcel is correct. Justify your answer.

1) Yes, Marcel is correct. The graph passes the vertical line test.

2) Yes, Marcel is correct. Every x-value on the graph has a y-value.

3) No, Marcel is not correct. $x = 2$ is paired with two y-values, 3 and −1, so the graph fails the vertical line test.

4) No, Marcel is not correct. The graph has a V shape, so it cannot be a function.

STATION

2

UNIT 5 REVIEW #2

Which relation is a function?

Which relation is a function?

1) {(2, 1), (3, 1), (4, 1), (5, 1)}

2) {(1, 2), (1, 3), (1, 4), (1, 5)}

3) {(2, 3), (3, 2), (4, 2), (2, 4)}

4) {(1, 6), (2, 8), (3, 9), (3, 12)}

STATION

3

UNIT 5 REVIEW #2

Evaluate a function

If $f(x) = \frac{\sqrt{2x + 3}}{6x - 5}$, then $f(\frac{1}{2}) =$

a) 1

b) −2

c) −1

d) $-\frac{13}{3}$

STATION

4

UNIT 5 REVIEW #2

Evaluate a function

A function is defined as $K(x) = 2x^2 - 5x + 3$. The value of $K(-3)$ is

a) 54

b) 36

c) 0

d) −18

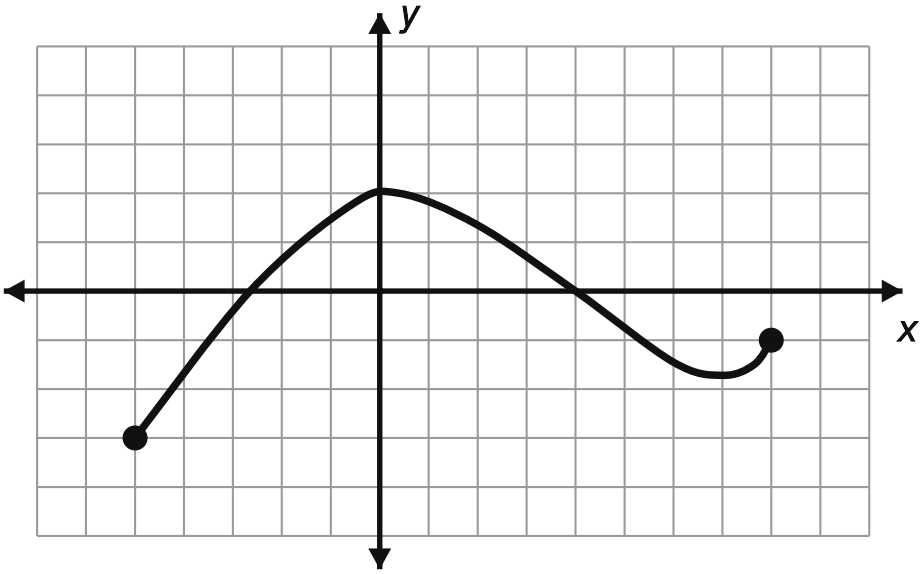
STATION

5

UNIT 5 REVIEW #2

Reading a graph

The graph below represents the function $y = f(x)$.



a) State the domain and range of this function.

A) Domain: $-5 \le x \le 8$
Range: $-3 \le y \le -1$

B) Domain: $-5 \le x \le 8$
Range: $-3 \le y \le 2$

C) Domain: $-3 \le x \le 2$
Range: $-5 \le y \le 8$

D) Domain: $-10 \le x \le 10$
Range: $-10 \le y \le 10$

b) What is the value of x when $f(x) = -3$?

c) What is $f(4)$?

STATION

6

UNIT 5 REVIEW #2

Subtracting polynomials

What is $(x - 4)^2$ subtracted from $(x^2 + 3x - 4)$?

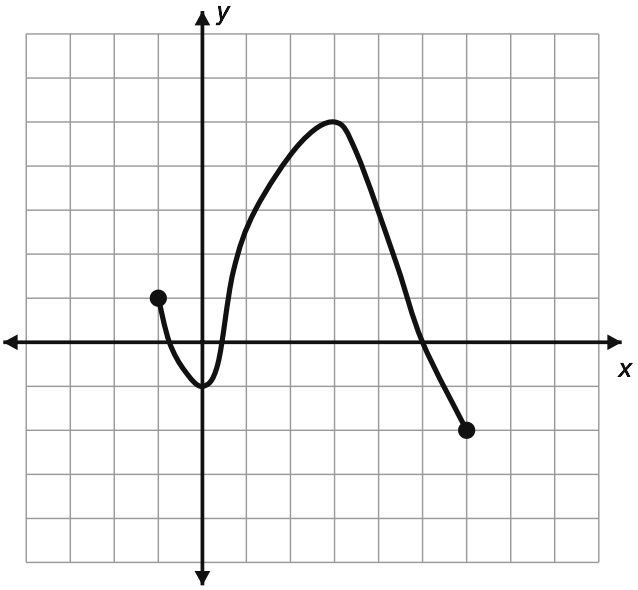
STATION

7

UNIT 5 REVIEW #2

Domain from a graph

What is the domain of the function shown below?



1) $-1 \le x \le 6$

2) $-1 \le y \le 6$

3) $-2 \le x \le 5$

4) $-2 \le y \le 5$

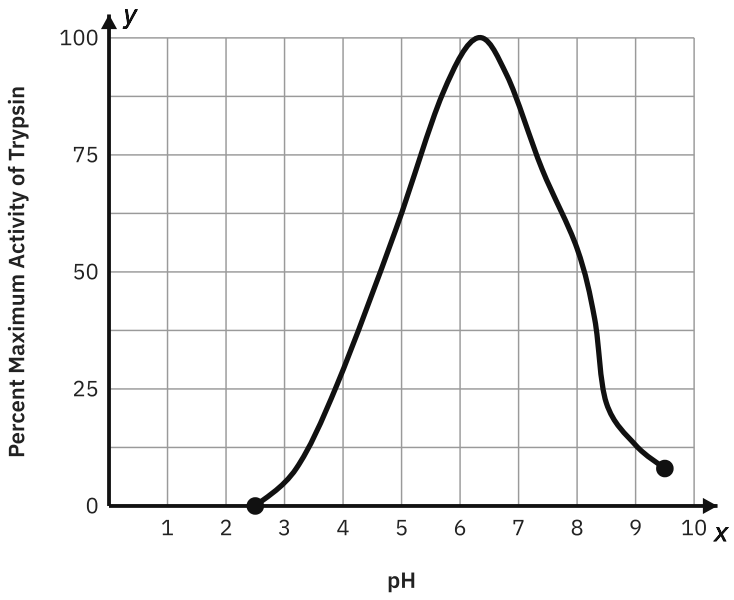
STATION

8

UNIT 5 REVIEW #2

Range of data

Data collected during an experiment are shown in the accompanying graph.



What is the range of this set of data?

1) $2.5 \leq y \leq 9.5$

2) $2.5 \leq x \leq 9.5$

3) $0 \leq y \leq 100$

4) $1 \leq x \leq 10$

STATION

9

UNIT 5 REVIEW #2

Terms from a formula

Find the first 5 terms given the formula

$f(n) = 3(n - 1) + 4$

STATION

10

UNIT 5 REVIEW #2

Terms from a formula

Find the first 5 terms given the formula $f(n) = 2(5)^{n-1}$

STATION

11

UNIT 5 REVIEW #2

Sequence: 1, 2, 4, 8, ...

Given the following: 1, 2, 4, 8, 16, 32, ...

a) Is the pattern Arithmetic, Geometric, or Neither?

A) Arithmetic **B)** Geometric **C)** Neither

b) Write a rule that represents the *n*th term of the sequence.

c) Find the 10th term of the sequence.

STATION

12

UNIT 5 REVIEW #2

Sequence: 4, 7, 10, ...

Given the following: 4, 7, 10, 13, 16, ...

a) Is the pattern Arithmetic, Geometric, or Neither?

A) Arithmetic **B)** Geometric **C)** Neither

b) Write a rule that represents the *n*th term of the sequence.

c) Find the 12th term of the sequence.

STATION 13

UNIT 5 REVIEW #2

Build an arithmetic sequence

Write an arithmetic sequence in which $a_1 = 3$ and $a_4 = 15$. Find the 13th term.

a) Write a rule for the nth term.

b) Find the 13th term.

STATION

14

UNIT 5 REVIEW #2

Geometric sequence with fractions

The terms of a geometric sequence are given below.

$$1, \quad \frac{1}{3}, \quad \frac{1}{9}, \quad \frac{1}{27}, \quad \ldots$$

a) Write a rule for the nth term of this geometric sequence.

b) Determine the 7th term.

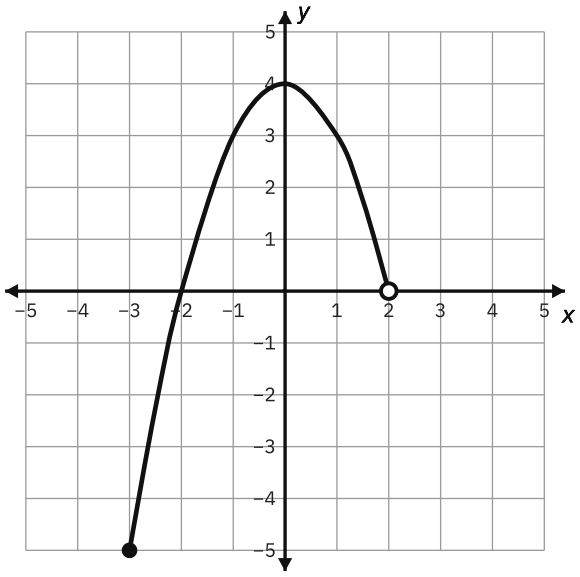
STATION

15

UNIT 5 REVIEW #2

Domain and range

State the domain and range of the function.



A) Domain: $-3 \le x \le 2$
Range: $-5 \le y \le 4$

B) Domain: $-5 \le x \le 4$
Range: $-3 \le y < 2$

C) Domain: $-3 \le x < 2$
Range: $-5 \le y \le 4$

D) Domain: $-3 \le x < 2$
Range: $-5 \le y \le 0$

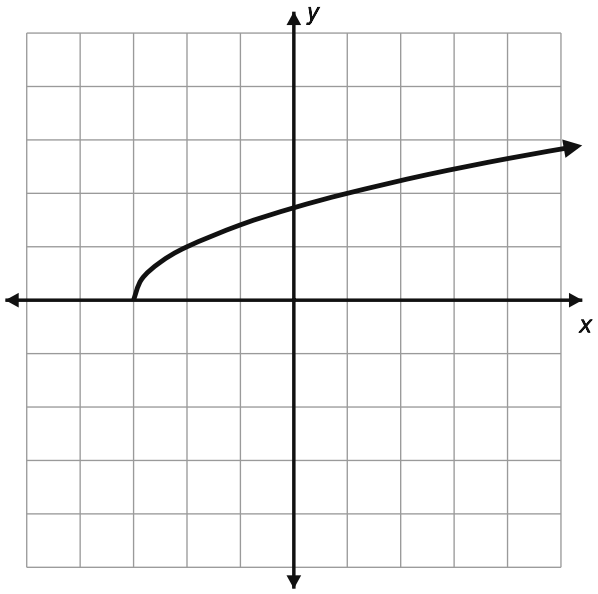
STATION

16

UNIT 5 REVIEW #2

Domain and range

State the domain and range of the function.



A) Domain: $x \geq 0$
Range: $y \geq -3$

B) Domain: $-3 \leq x \leq 5$
Range: $0 \leq y \leq 3$

C) Domain: all real numbers
Range: $y \geq 0$

D) Domain: $x \geq -3$
Range: $y \geq 0$

Another way to write it
Scan me

STATION

17

UNIT 5 REVIEW #2

Common ratio of –2

A geometric sequence with a common ratio of –2 is

a) 6, 4, 2, 0, ...

b) 5, 7, 9, 11, ...

c) –3, 6, –12, 24, ...

d) –10, –8, –6, –4, ...

STATION

18

UNIT 5 REVIEW #2

Solve for x

If $f(x) = \frac{5x - 9}{2}$, algebraically determine the value of x when $f(x) = 8$.

Show every step.